1 Solution — GIAM 1.7 Problem 6

1.1 Problem

A function $f(x)$ is *invertible* if there is another function $g(x)$ with $g(f(x)) = x$ for all x (the inverse is usually written $f^{-1}(x)$). Suppose a function is presented to you as a relation — a set of pairs. How can you tell whether it is invertible? How can you produce the inverse as a set of ordered pairs?

1.2 Answer

Suppose the function is the pair-set $f = \{(x_1, y_1), (x_2, y_2), \ldots\}$, where each first coordinate is an input and each second coordinate its output.

Test for invertibility. The function is invertible **if and only if all the second coordinates (outputs) are distinct** — no two pairs share the same output value:

$$\boxed{f \text{ is invertible} \iff (x_i, y) \in f \text{ and } (x_j, y) \in f \;\Rightarrow\; x_i = x_j.}$$

Equivalently: f is **one-to-one** (injective).

Producing the inverse. Simply **swap the two coordinates of every pair**:

$$\boxed{f^{-1} = \{\, (y, x) \mid (x, y) \in f \,\}.}$$

When the outputs were all distinct, this swapped set is again a genuine function, and it is the inverse f^{-1}.

1.3 Why This Works

A set of pairs is a *function* when each first coordinate appears **once** (each input has a single output). When we swap every pair, the old *outputs* become the new *inputs*. So the swapped set is a function exactly when the old outputs were all distinct — otherwise some new input would carry two different outputs, and “function” fails. That is precisely the one-to-one condition.

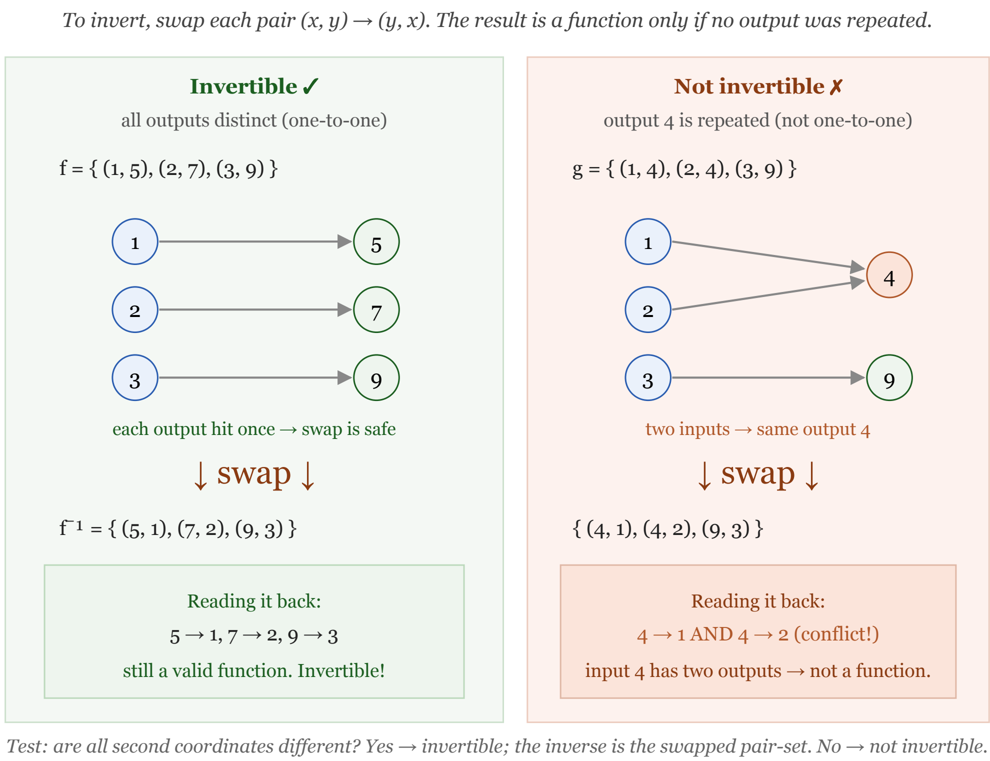

Figure 1.1: Left, an invertible example: f = {(1,5),(2,7),(3,9)} has all outputs 5,7,9 distinct, so swapping gives f-inverse = {(5,1),(7,2),(9,3)}, which reads 5→1, 7→2, 9→3 — still a valid function. Right, a non-invertible example: g = {(1,4),(2,4),(3,9)} repeats the output 4 (both 1 and 2 map to 4), so swapping gives {(4,1),(4,2),(9,3)}, where the input 4 would map to both 1 and 2 — not a function. The test: are all second coordinates different? Yes means invertible with inverse the swapped set; no means not invertible.

1.4 A Worked Pair of Examples

Invertible. $f = \{(1,5),(2,7),(3,9)\}$. Outputs $5,7,9$ are distinct, so f is invertible and

$$f^{-1} = \{(5,1),(7,2),(9,3)\}.$$

Check: $f^{-1}(f(2)) = f^{-1}(7) = 2$ ✓.

Not invertible. $g = \{(1,4),(2,4),(3,9)\}$. The output 4 is repeated (both 1 and 2 map to it). Swapping gives $\{(4,1),(4,2),(9,3)\}$, in which the input 4 is paired with *both* 1 and 2 — not a function. So g has no inverse: given the output 4, you cannot recover which input produced it.

1.5 Remark — The Set-Form of the Horizontal Line Test

The "all outputs distinct" rule is exactly the **horizontal line test** rewritten for pairs. Graphically, a function is one-to-one when no horizontal line $y = c$ meets its graph twice; in pair language, "the graph meets $y = c$ twice" means two pairs share the second coordinate c. So checking for a repeated second coordinate *is* the horizontal line test. Compare Problem 2: there the *vertical* line test (no repeated **first** coordinate) decided whether a relation is a function at all; invertibility adds the *horizontal* test (no repeated **second** coordinate) on top.

1.6 Remark — Swapping Coordinates Is Reflecting Across $y = x$

Producing f^{-1} by swapping each (x, y) to (y, x) has a clean geometric meaning: it **reflects the graph across the line $y = x$**. That is why the graph of f^{-1} is always the mirror image of the graph of f in that diagonal — the picture-level version of the pair-swap. It also explains the defining property both ways round: not only $f^{-1}(f(x)) = x$, but also $f(f^{-1}(y)) = y$, since swapping twice returns the original pair. The inverse "undoes" f, and f "undoes" f^{-1}.

1.7 Remark — Injective, Surjective, and What "Invertible" Really Needs

Strictly, whether f^{-1} is a function on a *prescribed* codomain depends on two properties. **Injectivity** (distinct outputs) is what guarantees the swap has no input-conflict — this is the crux and the answer to the problem. **Surjectivity** (every codomain value is actually hit) governs whether f^{-1} is defined on *all* of the intended codomain or only on the range. When a function is presented merely as a list of pairs — with no codomain specified beyond "the outputs that occur" — the range *is* the codomain, so surjectivity is automatic and **injectivity alone decides invertibility**. This is why, for a bare set of pairs, the single check "are all second coordinates distinct?" is exactly right.

1.8 Remark — Connecting to $f(x) = x^2 \bmod 7$ (Problem 2)

The function of Problem 2, $f = \{(0,0),(1,1),(2,4),(3,2),(4,2),(5,4),(6,1)\}$, is a perfect illustration of a **non**-invertible function. Its outputs are not distinct — 2 and 5 both give 4, 3 and 4 both give 2, 1 and 6 both give 1 — so it fails the test. Swapping would produce pairs like $(4,2)$ and $(4,5)$, forcing the input 4 to two outputs. Concretely: knowing $x^2 \equiv 4 \pmod 7$ does not tell you whether $x = 2$ or $x = 5$. This is the reason square-root "functions" are multivalued, and why cryptographic schemes built on squaring modulo a number are hard to reverse — the many-to-one collapse destroys invertibility.